Notebook 6 — Chapter 6.3

Combinatorial & Two-Player Game Theory

21 worked examples — P/N positions, Nim and XOR, payoff matrices, minimax and mixed equilibria — calibrated for Stage 1/Stage 2 speed and rigor

Core Toolkit (read this first)

Normal play: players alternate, the player who cannot move *loses*. Label positions backwards from the end:

$$P: \text{ every move leads to an } N\text{-position}$$

$$N: \text{ some move leads to a } P\text{-position}$$

P = *previous* player wins (mover loses); N = *next* player wins. Terminal positions are P.
Nim: heaps $a_1, \dots, a_k$ with nim-sum $s = a_1 \oplus \cdots \oplus a_k$ (bitwise XOR):

$$P \iff s = 0. \qquad \text{If } s \neq 0: \text{ reduce a heap } a \text{ with } a \oplus s < a \text{ to } a \oplus s.$$

Matrix games (row player maximises A_{ij}): row guarantees $\underline{v} = \max_i \min_j A_{ij}$, column holds row to $\overline{v} = \min_j \max_i A_{ij}$, always $\underline{v} \le \overline{v}$. The **minimax theorem** gives equality with mixed strategies. For $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ without a saddle point (**indifference principle**):

$$p = \frac{d-c}{a+d-b-c}, \quad q = \frac{d-b}{a+d-b-c}, \quad v = \frac{ad-bc}{a+d-b-c}$$

A. Impartial Games: P- and N-Positions

Example 1 — The Subtraction Game $\{1, 2, 3\}$

A pile of n stones; each turn remove 1, 2 or 3 stones; the player who cannot move loses. Classify $n = 0, \dots, 12$, then find a winning first move from $n = 21$.
Solution. Build the table from the bottom: $n = 0$ is P; n is N iff $n-1$, $n-2$ or $n-3$ is P.

n	0	1	2	3	4	5	6	7	8	9	10	11	12
	P	N	N	N	P	N	N	N	P	N	N	N	P

From $n = 1, 2, 3$ you can reach 0; from 4 every move reaches $1, 2, 3$ (all N), so 4 is P. The pattern repeats: P iff $4 \mid n$.
For $n = 21$: $21 \equiv 1 \pmod 4$, so it is N; remove 1 to leave 20.
Strategy: whatever k the opponent removes, answer with $4 - k$.
⋆ *Analogy:* a see-saw that you keep balanced at multiples of four. After every pair of moves the pile drops by exactly 4, and you always arrive at the pivot first.

Example 2 — The Subtraction Game $\{1, 3, 4\}$

Moves remove 1, 3 or 4 stones. Find all P-positions, and say what to do from $n = 1000$ and who wins from $n = 100$.
Solution. Compute upward:

n	0	1	2	3	4	5	6
	P	N	P	N	N	N	N

n	7	8	9	10	11	12	13
	P	N	P	N	N	N	N

Key steps: $n = 2$ can only go to 1 (an N), so P. $n = 7$ goes to $6, 4, 3$, all N, so P. $n = 9$ goes to $8, 6, 5$, all N, so P. $n = 14$ repeats $n = 0$.

$$P\text{-positions}: \quad n \equiv 0 \text{ or } 2 \pmod 7$$

$100 \equiv 2 \pmod 7$: a P-position, so the *second* player wins. $1000 \equiv 6 \pmod 7$: remove 4 to reach $996 \equiv 2 \pmod 7$.
▶ *Speed trick:* for any subtraction game, P-positions are eventually periodic. Compute until the pattern of k consecutive entries repeats (k = max move).

Example 3 — Remove Any Power of Two

From a pile of n stones you may remove any power of two $(1, 2, 4, 8, \dots)$ stones. Find the P-positions and decide $n = 2026$.
Solution. Claim: $P \iff 3 \mid n$.
From a multiple of 3: $2^k \bmod 3 \in \{1, 2\}$, never 0, so every move leaves a non-multiple of 3.
From a non-multiple: if $n \equiv 1$ remove 1; if $n \equiv 2$ remove 2. Either way you reach a multiple of 3.
Since 0 is P, induction proves the claim. $2026 = 3 \cdot 675 + 1 \equiv 1$, so 2026 is N: remove 1.
⋆ *Analogy:* the moves are *keys that never fit the lock* $3 \mid n$: nobody can step onto a multiple of 3 from another multiple of 3, but from anywhere else a one- or two-stone key opens the lock.

Example 4 — Normal Play Versus Misère

In the $\{1, 2, 3\}$ game, now the player who takes the *last* stone **loses** (misère play). Find the losing positions and compare with Example 1 at $n = 22$.
Solution. Now position 1 is losing for the mover (forced to take the last stone). From $2, 3, 4$ you can reach 1, so they win; from 5 every move reaches $2, 3, 4$ (all wins for the next player), so 5 loses:

$$\text{losing for the mover}: \quad n \equiv 1 \pmod 4$$

At $n = 22 \equiv 2 \pmod 4$: remove 1 to leave 21. Under normal play (Example 1) the winning move from 22 is to remove 2, leaving 20.
▶ *Speed trick:* always read the last line of the rules. "Last mover wins" (P at $4k$) and "last mover loses" (P at $4k+1$) differ by exactly one stone.
⋆ *Analogy:* is the last slice of cake a prize or a chore? The convention just shifts the pivot by one.

Example 5 — Race to the Corner

A token sits at (x, y) with $x, y \ge 0$. A move shifts it *one* step left, *one* step down, or one step diagonally left-down. The player who moves it to $(0, 0)$ wins. Classify the positions and decide $(7, 5)$.
Solution. $(0, 0)$ has no moves, so it is P. Claim: $P \iff x, y$ both even.
Both even ⇒ every move fails: left or down changes one parity, diagonal changes both; the result always has an odd coordinate.
Otherwise a good move exists: (odd, odd): diagonal; (odd, even): left; (even, odd): down.
$(7, 5)$ is (odd, odd): step diagonally to $(6, 4)$.
⋆ *Analogy:* the board is a *chessboard of safe islands*: the even–even squares. Every step off an island lands on water, and from water one step always gets you back to an island.

Example 6 — Breaking the Chocolate Bar

An $m \times n$ chocolate bar of unit squares. A move breaks one piece along a grid line into two pieces. The player who cannot move (all pieces are single squares) loses. Who wins on 6×8 and on 5×7?
Solution. Each move increases the number of pieces by exactly one. You start with 1 piece and the game ends at mn pieces, so the game lasts exactly $mn - 1$ moves, *however anyone plays.*

$$6 \times 8: \ 47 \text{ moves (odd)} \Rightarrow \text{first player wins}$$

$$5 \times 7: \ 34 \text{ moves (even)} \Rightarrow \text{second player wins}$$

▶ *Speed trick:* before building a P/N table, look for an *invariant* that fixes the length of the game. Here every strategy is equally good.
⋆ *Analogy:* walking up a staircase of $mn - 1$ steps — you may stride or dawdle but the step count is fixed, so only its parity decides who arrives.

B. Nim and the XOR Nim-Sum

Example 7 — Computing a Nim-Sum and Finding Every Winning Move

Heaps $(3, 5, 7)$. Compute the nim-sum, decide who wins, and list *all* winning moves.
Solution. In binary: $3 = 011$, $5 = 101$, $7 = 111$. XOR each column (an even number of 1s gives 0):

$$011 \oplus 101 \oplus 111 = 001, \qquad s = 1 \neq 0 \ \Rightarrow \ N$$

A heap a can be reduced to $a \oplus s$ only if $a \oplus s < a$, i.e. if a has a 1 in the leading bit of s:

$$3 \oplus 1 = 2 < 3, \quad 5 \oplus 1 = 4 < 5, \quad 7 \oplus 1 = 6 < 7$$

So there are **three** winning moves, each removing one stone: $(2, 5, 7)$, $(3, 4, 7)$, $(3, 5, 6)$. Check $(2, 5, 7)$: $010 \oplus 101 \oplus 111 = 000$.
⋆ *Analogy:* XOR is a *column parity detector*. Each binary column must contain an even number of 1s; the position is a P exactly when all columns are even.

Example 8 — Why $(1, 2, 3)$ Is a P-Position

Show that the second player wins Nim on $(1, 2, 3)$, and play out a sample line.

Solution. $01 \oplus 10 \oplus 11 = 00$, so $s = 0$.
Every move breaks the balance: a move changes exactly one heap a to $a' \neq a$. The XOR of the other heaps is a (because the total was 0), so the new nim-sum is $a \oplus a' \neq 0$.
The second player can always restore it (Example 10). Sample line: you move $(1,2,3) \to (1,2,1)$, nim-sum 2; reply by removing the whole heap of size 2:

$$(1,2,1) \to (1,0,1), \qquad s = 0$$

Then the opponent must take a 1, and you take the other.
▶ *Speed trick:* $1 \oplus 2 \oplus 3 = 0$ is the shortest three-heap P-position. In general $a \oplus b \oplus (a \oplus b) = 0$.

Example 9 — A Four-Heap Position

Heaps $(5, 8, 11, 14)$. Find the winning moves.
Solution. In binary: 0101, 1000, 1011, 1110.

$$0101 \oplus 1000 = 1101, \quad \oplus 1011 = 0110, \quad \oplus 1110 = 1000$$

So $s = 8$, with leading bit 1000. Only heaps with that bit set can be reduced:

$$8 \oplus 8 = 0, \qquad 11 \oplus 8 = 3, \qquad 14 \oplus 8 = 6$$

The heap $5 = 0101$ has no 8-bit: $5 \oplus 8 = 13 > 5$, not allowed. Hence exactly three winning moves:

$$(5, 0, 11, 14), \quad (5, 8, 3, 14), \quad (5, 8, 11, 6)$$

Check $(5, 8, 11, 6)$: $0101 \oplus 1000 \oplus 1011 \oplus 0110 = 0000$.
⋆ *Analogy:* the leading bit of s is the *loudest wrong note*. You silence it by changing a heap that is currently playing that note, and the lower bits get fixed at the same time.

Example 10 — The Proof of Bouton's Theorem, by Example

Prove (i) from nim-sum 0 every move gives nim-sum $\neq 0$, and (ii) from nim-sum $\neq 0$ some move gives 0. Illustrate on $(12, 9, 6)$.
Solution. *(i)* Shown in Example 8.
(ii) Let $s \neq 0$ and let b be its leading 1-bit. Some heap a has bit b set (otherwise the column of b could not be odd). Replace a by $a \oplus s$. This agrees with a on all bits above b and has bit b cleared, so $a \oplus s < a$; the new nim-sum is $s \oplus s = 0$.
Illustration: $12 = 1100$, $9 = 1001$, $6 = 0110$:

$$1100 \oplus 1001 = 0101, \qquad 0101 \oplus 0110 = 0011 = s$$

Leading bit of $s = 3$ is the 2s place. Only $6 = 0110$ has it: $6 \oplus 3 = 5 < 6$. Move $(12, 9, 6) \to (12, 9, 5)$: $1100 \oplus 1001 \oplus 0101 = 0$. Heaps 12 and 9 have no 2-bit ($12 \oplus 3 = 15$, $9 \oplus 3 = 10$ would *increase* them), so this is the unique winning move.
Terminal position $(0, \ldots, 0)$ has $s = 0$ and is P; induction on total stones finishes the proof.
▶ *Speed trick:* to find a winning move, compute s, identify its leading bit, pick *any* heap with that bit, and replace it by $a \oplus s$.

Example 11 — Counting P-Positions

Three Nim heaps a, b, c are chosen independently with $a, b, c \in \{1, \ldots, 7\}$. Count the P-positions and find the probability that the *second* player wins. Repeat when heaps can also be empty ($\{0, \ldots, 7\}$).
Solution. $P \iff c = a \oplus b$. For $a, b \in \{1, \ldots, 7\}$, $a \oplus b \leq 7$ always, and $a \oplus b \geq 1$ iff $a \neq b$:

$$\#P = 7 \cdot 6 = 42, \qquad \mathbb{P}(P) = \frac{42}{343} = \frac{6}{49} \approx 0.122$$

With empty heaps allowed, a, b are free (64 choices) and $c = a \oplus b \in \{0, \ldots, 7\}$ is automatic:

$$\mathbb{P}(P) = \frac{64}{512} = \frac{1}{8}$$

▶ *Speed trick:* for XOR constraints, choose all but one variable freely; the last is forced. for $m \geq 2$ independent heaps uniform on $\{0, \ldots, 2^k - 1\}$, $\mathbb{P}(\text{XOR} = 0) = 2^{-k}$.
⋆ *Analogy:* the second player's winning set is a *thin diagonal slab* of the cube — one coordinate is locked to the other two.

Example 12 — When XOR Fails: Wythoff's Game

Two heaps. A move removes any positive number from *one* heap, or the *same* number from **both** heaps. List the first P-positions, show the XOR rule fails, and find the winning moves from $(5, 8)$.
Solution. $(0,0)$ is P. $(1,2)$: its moves reach $(0,2), (1,1), (1,0), (0,1)$, each of which can go to $(0,0)$, so all are N and $(1,2)$ is P. Continuing,

$$P: \ (0,0), \ (1,2), \ (3,5), \ (4,7), \ (6,10), \ (8,13), \ (9,15), \ \ldots$$

Rule: the k-th pair is $(a_k, a_k + k)$ with a_k the smallest integer not yet used. (Equivalently $(\lfloor k\varphi \rfloor, \lfloor k\varphi^2 \rfloor)$ with $\varphi = \frac{1+\sqrt{5}}{2}$.)
The XOR rule fails: $(1,2)$ has $1 \oplus 2 = 3 \neq 0$ but is P, while $(3,3)$ has XOR 0 but is N (take both).
From $(5,8)$ (5 and 8 are not a pair): either remove 5 from the big heap to get $(5,3) = (3,5)$, or remove 1 from *both* to get $(4,7)$.
⋆ *Analogy:* the extra diagonal move is like adding a *trapdoor* to the board. Nim's tidy XOR pattern breaks, and the golden ratio takes its place.

Example 13 — Sprague–Grundy: Every Game Is Secretly a Nim Heap

Define the Grundy value $G(n) = \text{mex}\{G(n-1), G(n-2), G(n-3)\}$ for the subtraction game $\{1, 2, 3\}$ (mex = smallest non-negative integer missing). A position is the sum of: a subtraction pile of 7 stones, a Nim heap of 5, and a Nim heap of 6 (a move acts on one component). Who wins? What if the pile has 8?
Solution. Compute G:

n	0	1	2	3	4	5	6	7	8
G	0	1	2	3	0	1	2	3	0

$G(n) = n \bmod 4$

The sum of games has Grundy value equal to the XOR of components. With pile 7: $G = 3$:

$$3 \oplus 5 \oplus 6 = 011 \oplus 101 \oplus 110 = 000 \ \Rightarrow \ P$$

The second player wins. With pile 8: $G = 0$, nim-sum $0 \oplus 5 \oplus 6 = 3 \neq 0$; fix it by taking $6 \to 6 \oplus 3 = 5$ (the heap $5 \to 5 \oplus 3 = 6$ would increase). Then $0 \oplus 5 \oplus 5 = 0$.
▶ *Speed trick:* to analyse a sum of unfamiliar games, replace each by its Grundy value and XOR. A single game is P iff $G = 0$.
⋆ *Analogy:* every impartial game wears a *Nim-heap costume*; G tells you which size of heap is hiding underneath.

Example 14 — Misère Nim

Heaps $(3, 4, 5)$, but now the player who takes the *last* stone **loses**. Find the winning move. Then classify $(1,1,1)$, $(1,1,2)$ and $(2,2)$.
Solution. *Rule:* play exactly as in normal Nim as long as some heap larger than 1 remains after your move. When your normal-Nim move would leave only heaps of size ≤ 1, instead leave an **odd** number of 1-heaps.
$(3,4,5)$: $011 \oplus 100 \oplus 101 = 010$, $s = 2$. Normal move $3 \to 1$ gives $(1,4,5)$, nim-sum 0, and heaps larger than 1 remain, so it is also the misère winning move. (A computer search over all moves confirms it is the unique winner.)
$(1,1,1)$: three ones; the mover takes one, the opponent one, and the mover takes the last and **loses**: P.
$(1,1,2)$: leave an odd number of ones by reducing $2 \to 1$: $(1,1,1)$. So N (the normal move $2 \to 0$ leaving $(1,1)$ would lose).
$(2,2)$: nim-sum 0 with big heaps present: P.
⋆ *Analogy:* misère Nim is Nim with a *last-minute twist*: the strategy is identical until the endgame of single stones, where parity of the remaining ones takes over.

C. Two-Player Zero-Sum Matrix Games

Example 15 — Iterated Dominance

Row player picks a row and Column player a column simultaneously; Column pays Row the entry:

$$A = \begin{pmatrix} 3 & 5 & 2 \\ 5 & 8 & 3 \\ 1 & 2 & 4 \end{pmatrix}$$

Eliminate dominated strategies and solve the game.
Solution. Row $R_2 = (5, 8, 3)$ beats $R_1 = (3, 5, 2)$ in every column, so **delete** R_1. In the remaining game (rows R_2, R_3) the Column player minimises. Column $C_1 = (5, 1)$ is entry-wise $\leq C_2 = (8, 2)$, so **delete** C_2:

$$\begin{pmatrix} 5 & 3 \\ 1 & 4 \end{pmatrix} \quad (R_2, R_3 \text{ vs } C_1, C_3)$$

No saddle point (row minima 3, 1, so $\underline{v} = 3$; column maxima 5, 4, so $\overline{v} = 4$). Use the 2×2 formulas with $a = 5, b = 3, c = 1, d = 4$, denominator $a + d - b - c = 5$:

$$p(R_2) = \tfrac{4-1}{5} = \tfrac{3}{5}, \quad q(C_1) = \tfrac{4-3}{5} = \tfrac{1}{5}, \quad v = \tfrac{20-3}{5} = \boxed{\tfrac{17}{5}}$$

Check the deleted strategies: against $q = (\frac{1}{5}, 0, \frac{4}{5})$, R_1 gets $\frac{3}{5} + \frac{8}{5} = \frac{11}{5} < \frac{17}{5}$; against $p = (0, \frac{3}{5}, \frac{2}{5})$, C_2 costs $\frac{24}{5} + \frac{4}{5} = \frac{28}{5} > \frac{17}{5}$.
⋆ *Analogy:* never play a move that a rival option beats in every scenario — like declining a worse seat on every train.

Example 16 — Saddle Points and the Minimax Inequality

$$A = \begin{pmatrix} 3 & 5 & 2 & 7 \\ 1 & 4 & 0 & 6 \\ 2 & 3 & 2 & 4 \end{pmatrix}$$

Find $\underline{v}$, $\overline{v}$, the saddle points and optimal strategies.
Solution. Row minima: 2, 0, 2, so

$$\underline{v} = \max_i \min_j A_{ij} = 2$$

Column maxima: 3, 5, 2, 7, so

$$\overline{v} = \min_j \max_i A_{ij} = 2$$

$\underline{v} = \overline{v}$, so the game has value $\boxed{2}$. A saddle point is an entry that is the *minimum of its row* and the *maximum of its column*. Entry $(R_1, C_3) = 2$ qualifies, and so does $(R_3, C_3) = 2$.
Optimal pure strategies: Row plays R_1 or R_3 (or any mix of the two), Column plays C_3. Value is unique even though saddle points are not.
▶ *Speed trick:* always compute $\underline{v}$ and $\overline{v}$ first. If they agree, you are done — no mixing needed. Always $\underline{v} \le \overline{v}$, because the player who commits first can only lose by it.
⋆ *Analogy:* a saddle lies *lowest along the horse's length and highest across it*; a ball placed there stays put, just as neither player wishes to deviate.

Example 17 — Solving a 2×2 Game by Indifference

$$A = \begin{pmatrix} 3 & -1 \\ -2 & 4 \end{pmatrix}$$

Find the optimal mixed strategies and the value.
Solution. Row minima $-1, -2$ give $\underline{v} = -1$; column maxima $3, 4$ give $\overline{v} = 3$. No saddle: mixing is needed.
Row mixes $(p, 1-p)$ so Column is **indifferent**:

$$\text{col 1}: \ 3p - 2(1-p) = 5p - 2, \qquad \text{col 2}: \ -p + 4(1-p) = 4 - 5p$$

$$5p - 2 = 4 - 5p \ \Rightarrow \ p = \tfrac{3}{5}, \qquad v = 5 \cdot \tfrac{3}{5} - 2 = \boxed{1}$$

Column mixes $(q, 1-q)$ so Row is indifferent:

$$\text{row 1}: \ 4q - 1, \quad \text{row 2}: \ 4 - 6q \ \Rightarrow \ q = \tfrac{1}{2}, \ v = 1$$

Formula check: $v = \frac{ad-bc}{a+d-b-c} = \frac{12-2}{10} = 1$.
⋆ *Analogy:* a pitcher who mixes fastballs and curveballs so the batter's expected result is the same whatever he guesses. Predictability is the only thing a mix must avoid.

Example 18 — The Odd/Even Fingers Game

Both players simultaneously show 1 or 2 fingers. If the total is **even**, Even wins that many points from Odd; if **odd**, Odd wins that many from Even. Find the optimal strategies and the value, and the cost of the naive $\frac{1}{2}$–$\frac{1}{2}$ mix.
Solution. Payoff to Even:

	1	2
1	+2	−3
2	−3	+4

$a = 2, b = -3, c = -3, d = 4$

Denominator $a + d - b - c = 12$:

$$p = \tfrac{4+3}{12} = \tfrac{7}{12} \text{ (show 1)}, \qquad q = \tfrac{7}{12}, \qquad v = \tfrac{8-9}{12} = \boxed{-\tfrac{1}{12}}$$

The game favours Odd by $\frac{1}{12}$ per round. Check: with $p = \frac{7}{12}$, col 1 gives $5p - 3 = -\frac{1}{12}$ and col 2 gives $4 - 7p = -\frac{1}{12}$.
Naive mix $p = \frac{1}{2}$: col 1 pays Even $-\frac{1}{2}$, col 2 pays $+\frac{1}{2}$. Odd picks col 1, so Even earns only $-\frac{1}{2}$, which is worse by $\frac{5}{12}$ per round.
▶ *Speed trick:* the optimal frequencies are *not* the naive ones — they equalise your opponent's outcomes, not yours.

Example 19 — A 2×3 Game by the Graphical Method

$$A = \begin{pmatrix} 1 & 3 & -2 \\ -1 & 0 & 4 \end{pmatrix}$$

Find the value and both players' optimal strategies.
Solution. Let Row play row 1 with probability p. Against each column Row's payoff is a line in p:

$$C_1: \ 2p - 1, \qquad C_2: \ 3p, \qquad C_3: \ 4 - 6p$$

Column picks the lowest line; Row maximises that lower envelope. The envelope peaks where C_1 (rising) meets C_3 (falling):

$$2p - 1 = 4 - 6p \ \Rightarrow \ p = \tfrac{5}{8}, \qquad v = 2 \cdot \tfrac{5}{8} - 1 = \boxed{\tfrac{1}{4}}$$

At $p = \frac{5}{8}$, C_2 gives $\frac{15}{8} > \frac{1}{4}$ so it is never used. For Column, mix only C_1, C_3 with $(q, 0, 1-q)$ so Row is indifferent:

$$q - 2(1-q) = -q + 4(1-q) \Rightarrow q = \tfrac{3}{4}$$

So Column plays $(\frac{3}{4}, 0, \frac{1}{4})$.
⋆ *Analogy:* three roads under a roof; the roof's lowest point is the best you can guarantee, and the walls only touch it at two roads.

Example 20 — Capstone: Honest or Bluffing?

Each player antes 1. Player 1 is dealt High or Low with probability $\frac{1}{2}$ each (secret). Player 1 may **give up** (losing the ante) or **bet** 1 more. If he bets, Player 2 may **fold** (Player 1 wins the ante) or **call** (Player 1 wins the pot iff his card is High). Strategies: Honest = bet only with High; Bluff = always bet. Find the optimal mixed strategies and the game value.
Solution. Payoffs to Player 1: bet with High: call +2, fold +1. Bet with Low: call −2, fold +1. Give up: −1. Average over the card:

$$\text{Honest/Call}: \ \tfrac{1}{2}(2) + \tfrac{1}{2}(-1) = \tfrac{1}{2} \qquad \text{Honest/Fold}: \ \tfrac{1}{2}(1) + \tfrac{1}{2}(-1) = 0$$

$$\text{Bluff/Call}: \ \tfrac{1}{2}(2) + \tfrac{1}{2}(-2) = 0 \qquad \text{Bluff/Fold}: \ \tfrac{1}{2}(1) + \tfrac{1}{2}(1) = 1$$

$$A = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$$

No saddle ($\underline{v} = 0$, $\overline{v} = \frac{1}{2}$). Denominator $a + d - b - c = \frac{3}{2}$:

$$p(\text{Honest}) = \tfrac{1-0}{3/2} = \tfrac{2}{3}, \quad q(\text{Call}) = \tfrac{1-0}{3/2} = \tfrac{2}{3}, \quad v = \tfrac{\frac{1}{2}\cdot 1 - 0}{3/2} = \boxed{\tfrac{1}{3}}$$

Interpretation: with a Low card Player 1 should bluff a third of the time, and Player 2 should call two times out of three. The informed player wins $\frac{1}{3}$ per hand on average.
⋆ *Analogy:* a bluff works only if it is *rare enough to be believed but frequent enough to be feared.* The indifference principle finds the exact balance point.

Example 21 — Capstone: Weighted Rock–Paper–Scissors

Paper beats Rock for 2, Scissors beats Paper for 3, Rock beats Scissors for 1; ties pay 0. Find the optimal mixed strategies and the value.
Solution. Payoffs to Row (rows and columns ordered R, P, S):

$$A = \begin{pmatrix} 0 & -2 & 1 \\ 2 & 0 & -3 \\ -1 & 3 & 0 \end{pmatrix}$$

A is skew-symmetric ($A^T = -A$), so the game is *symmetric*: the value is 0. Column plays q so that every row has expected payoff 0 (indifference over the full support):

$$-2q_2 + q_3 = 0, \qquad 2q_1 - 3q_3 = 0$$

$$q_3 = 2q_2, \quad q_1 = 3q_2 \ \Rightarrow \ q = \left(\tfrac{1}{2}, \tfrac{1}{6}, \tfrac{1}{3}\right)$$

The same vector is optimal for Row by symmetry. Check: playing pure Rock against it, Row gets $0 \cdot \frac{1}{2} + 2 \cdot \frac{1}{6} - 1 \cdot \frac{1}{3} = 0$. Frequencies are proportional to $(3, 1, 2)$.
Why not a third each? Against a uniform opponent, pure Rock, Paper and Scissors earn

$$\tfrac{1}{3}(0 - 2 + 1) = -\tfrac{1}{3}, \qquad \tfrac{1}{3}(2 + 0 - 3) = -\tfrac{1}{3}, \qquad \tfrac{1}{3}(-1 + 3 + 0) = +\tfrac{2}{3}$$

so pure Scissors gains $\frac{2}{3}$ per round: uniform play is exploitable.
▶ *Speed trick:* skew-symmetric payoff matrix means value 0 and identical optimal strategies for both players; solve $Aq = 0$, $\sum q = 1$.
⋆ *Analogy:* unequal prizes tilt the table. The cheap weapon is played most often, because the weapons that beat it are risky.